

ARTICLE

Mechanism of crosslinking in benzoyl peroxide-initiated functionalization of vinyltriethoxysilane onto polypropylene in the water medium

Hao Yuan  | Qing Shao | Fengchun Liang | Hongwei Shi | Wenbo Song

SINOPEC Beijing Research Institute of
Chemical Industry, Beijing, China

Correspondence

Hao Yuan and Wenbo Song, SINOPEC
Beijing Research Institute of Chemical
Industry, Beijing 100013, China.
Email: yuanhao.bjhy@sinopec.com
(H. Y.) and songwb.bjhy@sinopec.com
(W. S.)

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Abstract

Silane grafting and moisture crosslinking is one of the most common methods in the production of crosslinked polyolefin. A remarkable increase in the percentage of gel in the product was observed in benzoyl peroxide-mediated aqueous suspension grafting of vinyltriethoxysilane onto polypropylene. Aimed to explore the crosslinking reaction mechanism in this functionalization process, comprehensive experiments had been conducted and the grafted products were characterized by mass spectrometry, ^1H NMR, and ^{29}Si NMR spectroscopy. Finally, a novel crosslinking mechanism was proposed based on the characterization results.

KEYWORDS

crosslinking, grafting, radical polymerization

1 | INTRODUCTION

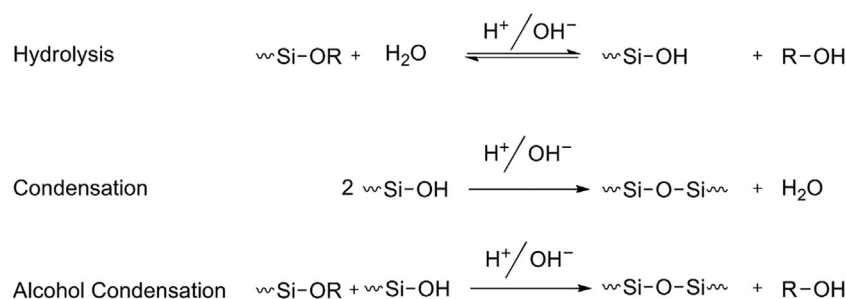
Crosslinked polyolefin is a value-added material practiced in applications such as wire and cable insulation materials and plastic pipes.^[1,2] There are three traditional technologies: peroxide-crosslinking,^[3] radiation-crosslinking,^[4] and silane-crosslinking. Therein, silane-crosslinking is achieved by grafting vinyl silane (VS) onto the polyethylene (PE) chain with a small amount of peroxide. Silane alkoxy groups are first to be introduced onto polyolefin backbone by peroxide-initiated grafting of unsaturated silane, then transformed to siloxane ($-\text{Si}-\text{O}-\text{Si}-$) through catalytically hydrolytic condensation using water or moisture (Scheme 1).^[5]

Compared with the other two methods, silane-crosslinking has advantages of low capital investment and favorable properties of the processed materials.^[6]

In the case of polypropylene (PP), predominant macro-radical fragmentation during radiation and peroxide modification^[7] means silane grafting and moisture crosslinking have a greater advantage in the production of crosslinked PP.^[8]

The conventional preparation route of silane-modified PP is peroxide-initiated functionalization in the melt by a twin-screw extruder. However, due to the reaction temperature (over 180°C), macro-radicals are prone to chain scission, which negatively affects graft efficiency and works against the effort to achieve a high degree of crosslinking in the subsequent stage.^[9]

Solid-state modification^[10,11] with mild reaction condition and a few deleterious reactions has emerged as an innovative way of polyolefin functionalization. Although various monomers have been successfully grafted onto polyolefin, reports of silane grafting by solid-state modification are rare. Rätzsch et al.^[12] studied irradiation grafting of monomers with various structures onto PE and PP and found out that graft degree of vinyl alkoxy silanes turned to be low, while VS did not tend to form long polymeric chains in grafting processes especially at room temperature and low dose rates because of high transfer rate. Jain^[13] reported the synthesis of PP/silica nanocomposites by combining solid-state modification of PP with vinyltriethoxysilane (VTES) and sol-gel



SCHEME 1 Reaction equations for hydrolysis, condensation, and alcohol condensation of alkoxy silane

technology with the idea that grafted monomeric VTES could be incorporated in the silica particles during the sol-gel reaction to control the matrix filler adhesion.

Suspension grafting^[14,15] is actually a special solid-state modification. The preferred pristine PP was synthesized by reactor granule technology^[16] with a very important characteristic of controlled high porosity (from 10 to more than 40 vol%). In the grafting reaction, the initiator and monomer first diffuse into the porous regions of PP, then further into the amorphous phase, and finally, the radical polymerization can be realized only in the amorphous areas of PP granules at the reaction temperature.^[10] The adsorption of monomer depends on PP porosity and the solubility of monomer in PP.^[17] Each PP granule acts as an independent reactor sealed by water as heat transfer and dispersing medium, which can enhance reaction efficiency and reduce low-molecular-weight by-product.

We have performed the functionalization of VTES onto PP in suspension and observed a considerable amount of gel in the product when benzoyl peroxide (BPO) was used as an initiator. To the best of our knowledge, no such observation has been reported before. In this article, the mechanism of gel generation in the grafting process is further studied by a series of experiments and characterization of grafted products.

2 | EXPERIMENTAL

2.1 | Materials

T30S grade PP homopolymer with a melt flow rate of 3.8 dg/min (ASTM D 1238, 230°C, and 2.16 kg), supplied in powder form by Maoming Petrochemical Company, was chosen as the base polymer for the current study. VTES, BPO (wetted with ~25%wt H₂O), lauroyl peroxide (LPO), benzoic acid, and acetone were obtained from Beijing J&K Technology Co. (Beijing, China). The water used in this work was distilled followed by deionization. All the chemicals and

polymers were used as purchased without any further purification or modification.

2.2 | Sample preparation

The grafting process was conducted in a 1 L four-necked round-bottom flask equipped with a twin-blade mechanical stirrer operated at 200 rpm, a condenser tube, a nitrogen-purging inlet, and a thermometer. About 200 g of T30S spheres were charged into the reactor and stirred under a flow of nitrogen to remove oxygen. BPO (0.5–2 g) was dissolved in 30 ml acetone, and then the solution was dropped in the flask for 10 min and mixed with T30S particles for 30 min. The reactor was first heated to 48°C and kept for 10 min to remove the acetone, and then heated further to 90°C. Twenty grams of VTES and about 400 ml of de-ionized water (ultrasonic deoxygenated, heated to 90°C) were added successively into the reactor, and then the graft polymerization proceeded at 90°C for 4 hr. All steps of reaction were carried out under nitrogen flow to have inert atmospheric conditions. The modified PP spheres were recovered by filtration and dried at 75°C under vacuum to reach a constant weight, then stored in airtight desiccators to avoid absorption of moisture. The unpurified product was in reality a mixture of grafted PP, unreacted PP, unreacted monomer, and homopolymer of VTES. About 2–4 g of products were refined by Soxhlet extraction using acetone for 24 hr and then dried to obtain a pure grafted sample.

2.3 | Grafting degree determination

The grafting degree (GD) was calculated as per the following equation:

$$\text{GD} = \frac{w_1 - w_0}{w_1} \times 100\%$$

where w_0 is the weight of pristine PP spheres and w_1 is the weight of grafted PP.

In this work, GD indicates the content of VTES in grafted products.

2.4 | Melt mass flow rate of T30S-g-VTES

The test sample was prepared with 10 g products mixed with 0.12 g antioxidants (168: Irganox 1010 = 1:1), and melt mass flow rate (MFR) was measured at 230°C with a load of 2.16 kg according to ASTM D1238, using a CEAST7026 melt indexer.

2.5 | Gel content measurement

The degree of crosslinking could be expressed by gel content measured according to standard ASTM D2765-2006. Approximately 0.3 g of specimens were packaged in a holder of 120-mesh stainless steel pouch and extracted with refluxing xylene containing 1% of antioxidant for 12 hr using a Soxhlet extractor. Samples were dried in a vacuum oven at 120°C for 24 hr. Gel content was calculated as follows:

$$\text{Gel}\% = \frac{w_2 - w_0}{w_1 - w_0} \times 100\%$$

where w_0 is the weight of pouch, w_1 is the weight of sample and pouch before extraction, and w_2 is the weight of sample and pouch after extraction and drying.

2.6 | Mass spectrometry

The composition of the aqueous solution after reaction and extraction solution was detected by Micromass Quattro Premier LC/MS/MS.

2.7 | Nuclear magnetic resonance spectroscopy

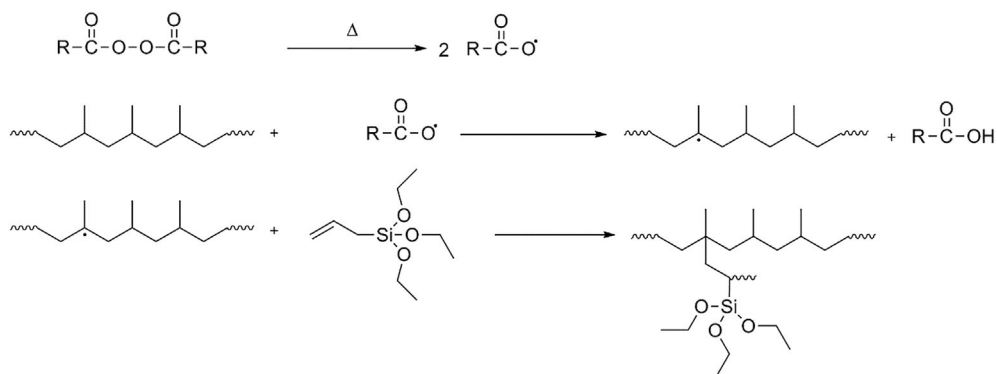
Agilent 400-MR DD2 nuclear magnetic resonance (NMR) spectrometer was used for solution ^1H NMR spectra to determine whether the silane has hydrolyzed to silanol during grafting. Samples were prepared by dissolving in deuterium *o*-dichlorobenzene. All the measurements were taken at 110°C with a spectral frequency of 400 MHz.

^{29}Si NMR spectroscopy was carried out on Agilent 600 MHz PremiumCompact⁺ spectrometer operating at a ^{29}Si NMR frequency of 119.169 MHz and equipped with a 4-mm magic-angle spinning probe head, and the sample rotation rate was 8 kHz. T^x represents silicon species, where x (=0, 1, 2, and 3) refers to the number of surrounding Si—O—Si bonds.^[18]

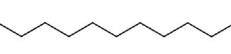
During functionalization, a part of the monomer was grafted onto PP while the other part might homopolymerize to PVTES, which eventually linked to the grafted product by condensation reaction. In this case, both ^1H NMR spectra and ^{29}Si NMR spectra were performed on unpurified samples to reveal the molecular structure of silane products.

2.8 | Headspace-gas chromatography-mass spectrometry

Headspace-gas chromatography-mass spectrometry (HS-GC-MS) was carried out on an Agilent 7,697-7890B-5,977 spectrometer. After dried with CaCl_2 for 24 hr, 2 g of specimens were placed in HS vial and incubated at 120°C for 60 min to release volatile substances. Furthermore, 1 ml of gaseous sample was directly taken out from the headspace of the vial and characterized by GC/MS.



SCHEME 2 Main reaction mechanism of peroxide-initiated grafting of vinyltriethoxysilane (VTES) on polypropylene (PP)

where R represents  or 

Run	1	2	3	4	5	6	7 ^a
Initiator concentration (wt%) ^b	2.50	3.75	5.00	6.25	7.50	10.00	12.34
GD (%) ^c	1.66	1.54	2.11	2.42	2.38	2.48	3.53
MFR (dg/min)	3.43	2.18	1.30	0.94	0.67	0.05	2.85
Gel content of powder (%) ^d	n.d.	n.d.	0.06	n.d.	0.87	2.13	n.d.
Gel content of extrudate ^{d,e} (%)	0.49	0.51	0.69	3.41	6.62	34.22	n.d.

TABLE 1 Results of T30S-g-VTES with different initiator concentrations

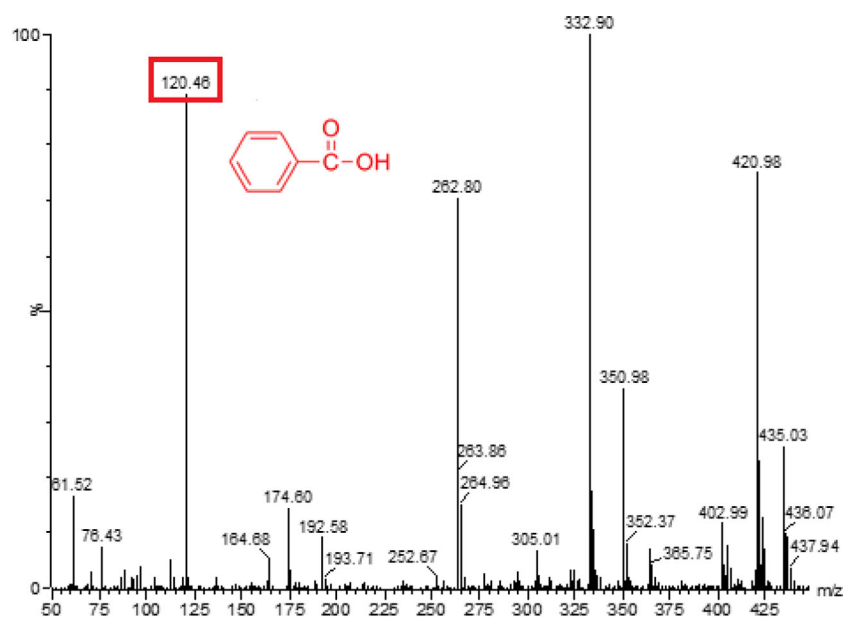
^aLauroyl peroxide (LPO) was used as an initiator.

^bWeight ratio of initiator to monomer $\times 100\%$.

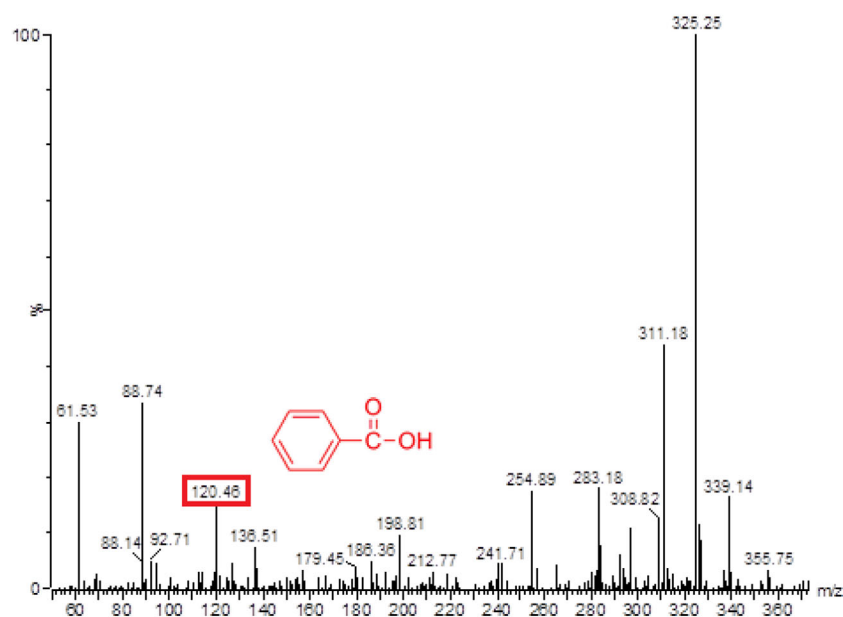
^cExperimental error: $\pm 0.1\%$.

^dExperimental error: $\pm 0.1\%$.

^eSpecimens extruded by the mass flow rate (MFR) indexer.



(a) Aqueous solution



(b) Extraction solution

FIGURE 1 Benzoic acid detected in Sample 6 by mass spectrometry (MS): (a) aqueous solution, (b) extraction solution [Color figure can be viewed at wileyonlinelibrary.com]

3 | RESULTS AND DISCUSSION

3.1 | Graft polymerization of VTES on T30S

The reaction mechanism of the grafting process is expressed in Scheme 2. In Runs 1–6, BPO was used as the initiator, the influence of initiator concentration on GD, MFR, and gel content was investigated. Run 7 was carried out as a controlled experiment to Run 6 with the only BPO replaced by the same molar of LPO. The results are shown in Table 1.

From Run 1 to Run 6, GD first raises up then reaches a constant value of 2.42% at initiator concentration of 6.25%. It means that GD can benefit by increasing initiator concentration but be limited by the amount of VTES

adsorbed in PP particles. The increase in initiator concentration causes a decrease in MFR and an increase in gel content. The gel content of powder remains at low level over the range of initiator concentration examined, the considerable increment in extrudate is caused by heat treatment in MFR measurement. In Runs 4–6, although GD shows little difference, the gel content of extrudate still increases with raising initiator concentration and reaches a maximum at the initiator concentration of 10%. As PP is obviously prone to degrade under this condition, insoluble gel in product results from silane crosslinking. Moreover, enhancement in both initiator concentration and temperature is likely to favor the crosslinking efficiency.

Compared with Run 6, little gel is detected despite higher GD in Run 7. Hence, it is clear that gel is

FIGURE 2 ^1H NMR spectrum of Sample 6 powder [Color figure can be viewed at wileyonlinelibrary.com]

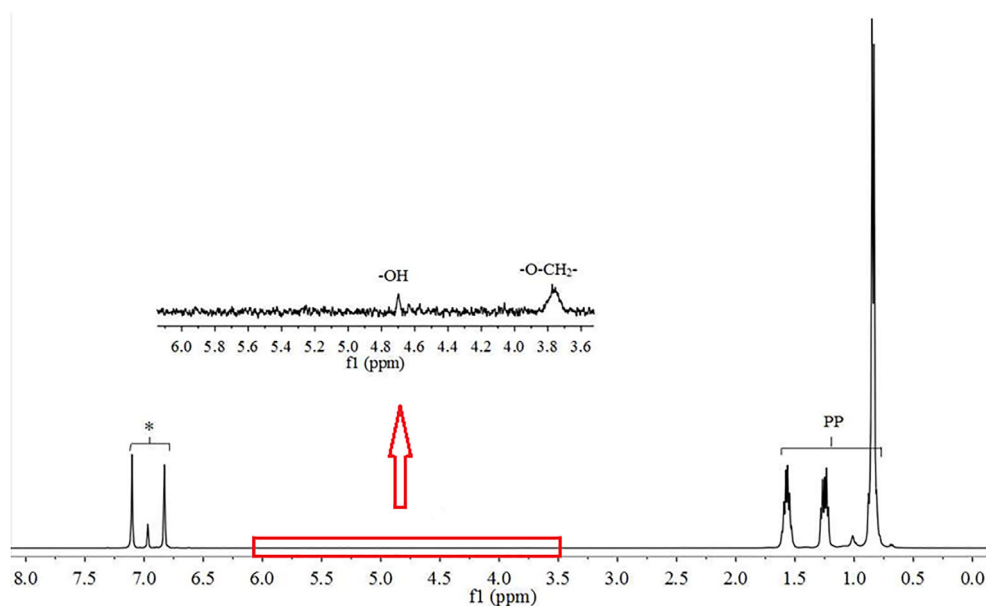
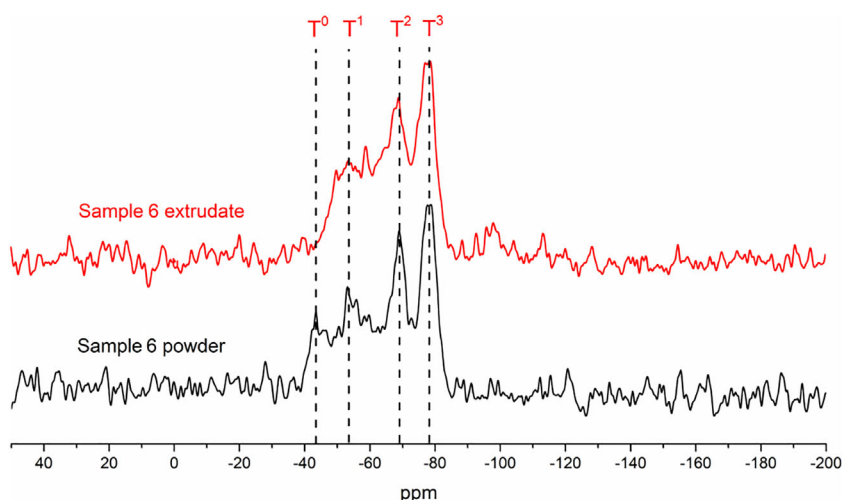


FIGURE 3 ^{29}Si NMR spectrum of Sample 6 [Color figure can be viewed at wileyonlinelibrary.com]



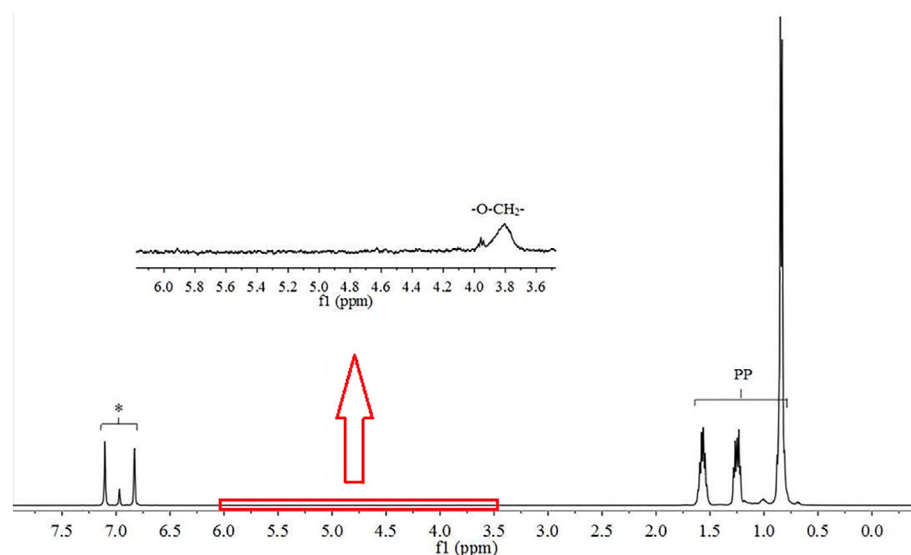


FIGURE 4 ^1H NMR spectrum of Sample 7 powder [Color figure can be viewed at wileyonlinelibrary.com]

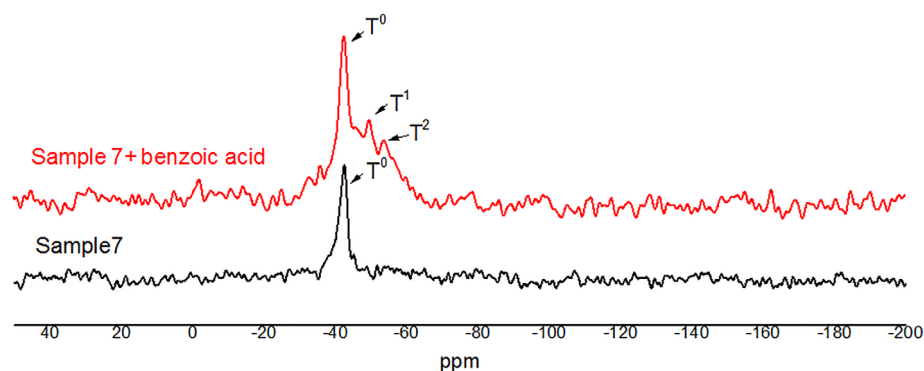


FIGURE 5 ^{29}Si NMR spectrum of Sample 7 extrudate before and after mixing benzoic acid [Color figure can be viewed at wileyonlinelibrary.com]

produced only in BPO-initiated modification; meanwhile, the crosslinking rate appears to be positive relative to BPO concentration.

As well known, spontaneous hydrolysis and condensation reactions of silane are reversible and proceed extremely slowly without the catalyst. In view of the reaction mechanism shown in Scheme 2, BPO first decomposes into primary radicals—benzoyloxy, which abstract hydrogen atoms from PP to convert to benzoic acid, which acts as a strong catalyst in the following silane crosslinking.^[19] The existence of benzoic acid in both aqueous solution and Sample 6 was also confirmed by MS in Figure 1.

To prove the determinant of benzoic acid in the production of gel, Sample 7 mixed with a certain amount of benzoic acid was extruded by the melt indexer. As expected, a decrease of MFR from 2.85 to 0.9 dg/10 min as well as a gel content of 5.6% was observed. This result with experimental data in Table 1, together gives further proof for the significant catalytic effect of benzoic acid on the crosslinking of silane.

In the following, the structures of Sample 6 and Sample 7 were analyzed by the NMR spectrometer in order to study the crosslinking mechanism in the grafting process.

Figure 2 shows the ^1H NMR spectrum of Sample 6 powder. The peak at 3.76 ppm (the inset shows the region of interest) is assigned to be the resonance of protons of ethoxy groups attached to the Si atom. Taking into account the equations shown in Scheme 1, the hydrolysis of ethoxysilane during modification has been borne out by resonances of hydroxy appeared at 4.70 ppm.

As shown in Figure 3, the signals of T^1 , T^2 , and T^3 observed in the ^{29}Si NMR spectrum confirm the formation of siloxane. Therefore, the NMR data of Sample 6 evidence that silane crosslinking in grafting does proceed via the reaction route shown in Scheme 1.

In the spectra of extrudate shown above, the disappearance of T^0 indicates the remaining $\text{Si}-\text{O}-\text{C}$ (H) transformed further to $\text{Si}-\text{O}-\text{Si}$ by heat treatment in MFR measurement. It is explained why gel content increased from 2.1% in powder to 34.2% in the extrudate. Compared with monofunctional silane, hydrolysis of

SCHEME 3 Generation of siloxane from the reaction between ethoxysilane and benzoic acid

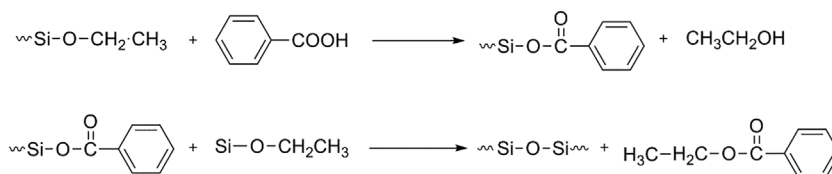
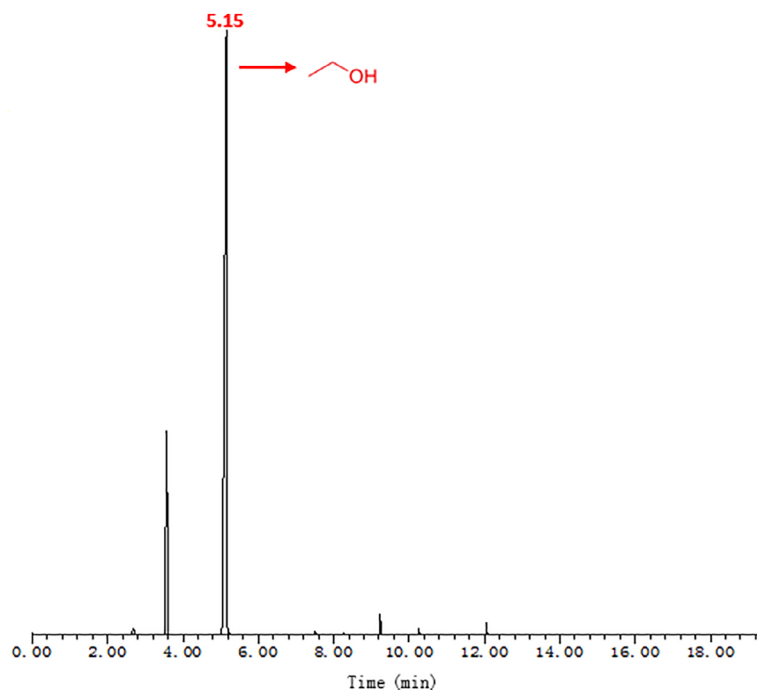


FIGURE 6 Gas chromatography (GC) spectrum of powder of mixture of Sample 7 and benzoic acid [Color figure can be viewed at wileyonlinelibrary.com]



trifunctional silane proceeds much slower, as the removal of alkoxy groups on the silicon atoms occurs not at once but in steps with each succeeding step slower than the former.^[20] Thus, the low conversion of ethoxysilane to siloxane contributes to a small amount of network in powder during functionalization. Through MFR measurement, gel content has raised substantially in view of the strong impact of temperature on crosslinking efficiency.^[20]

In the case of LPO, the decomposition product is dodecyl acid, a weaker acid than benzoic acid, which also means less catalytically active in silane crosslinking.^[21] Absence of —OH group signals in the ¹H spectrum (Figure 4) as well as only T⁰ peak detected in the ²⁹Si spectrum (Figure 5) suppose that the rate of hydrolysis and condensation of alkoxy silane are negligible in response to the reaction condition.

According to Figure 5, after mixing benzoic acid, the appearance of T¹ and T² in the ²⁹Si NMR spectrum indicates that benzoic acid can enable a direct conversion of silane to siloxane without hydrolysis. This consequence

appears to be against the traditional mechanism of silane crosslinking. This formation of Si—O—Si is potentially attributed to the direct reaction between benzoic acid and silane ethoxy (Scheme 3).

Furthermore, the detection of the important intermediate-ethyl alcohol in the mixture by HS-GC-MS (Figure 6) stands as another powerful proof for the reaction mechanism above.

4 | CONCLUSION

We have studied the mechanism of crosslinking reaction in BPO-initiated VTES grafting PP in suspension. The widely accepted silane crosslinking mechanism has been proved by NMR, besides a diverse mechanism has been proposed.

Overall, it is benzoic acid, the decomposition of BPO, which played a key role in the generation of Si—O—Si. Benzoic acid can catalyze the hydrolysis of ethoxysilane as well as condensation of silanol in suspension functionalization. On the other hand, siloxane can also be

generated by the direct reaction between benzoic acid and Si—O—R groups. A novel mechanism is seldom mentioned in previous researches in the field of silane-crosslinked polyolefin. Besides, in either case, the rate of crosslinking appears to be relatively low at reaction temperature and enhanced notably by elevating temperature.

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ORCID

Hao Yuan  <https://orcid.org/0000-0002-7137-0530>

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